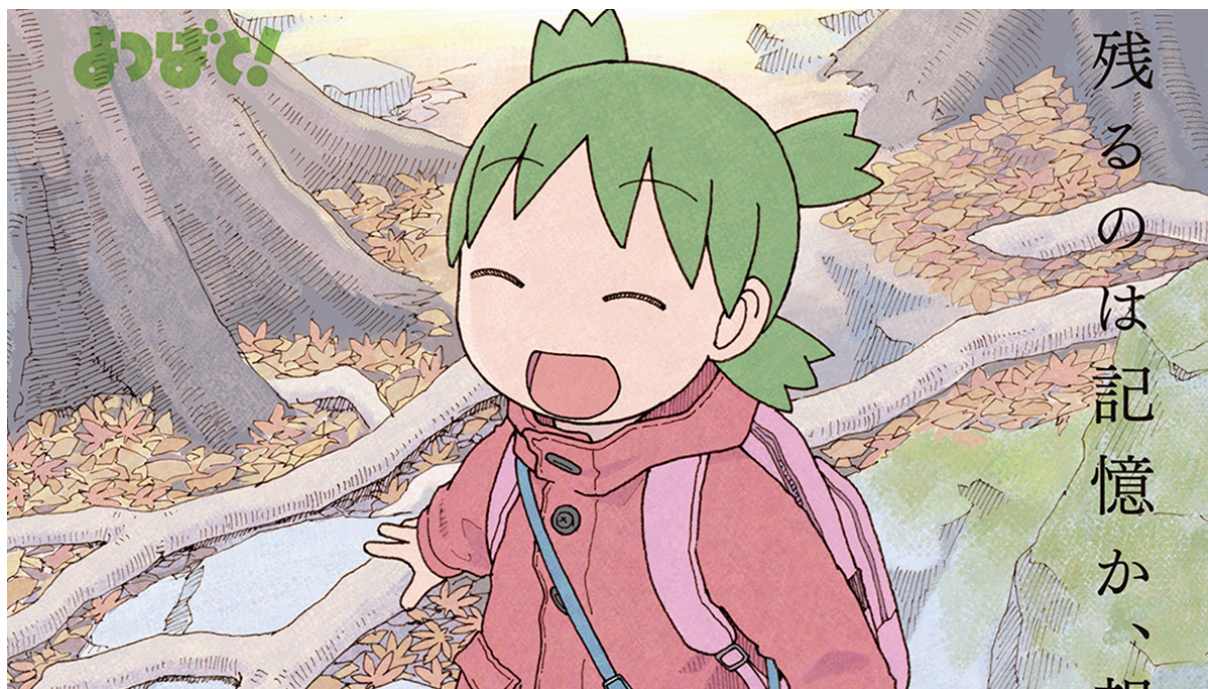


MATH 284 Notes

Linear Algebra

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Yotsuba from Yotsuba&!

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1 Linear Equations in Linear Algebra

1.1 Systems of Linear Equations

- A **linear equation** with the variables $x_1, x_2, \dots, x_n$ and coefficients $a_1, a_2, \dots, a_n$ can be written as follows:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

- A **system of linear equations** is a collection of numerous linear equations all using the same variables.

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = k_1$$

$$b_1x_1 + b_2x_2 + \dots + b_nx_n = k_2$$

- The solutions of a system of linear equations have three variants:
 - **No solution.** In $\mathbb{R}^2$, the linear equations would be parallel, non-overlapping lines.
 - **One solution.** In $\mathbb{R}^2$, the linear equations would be nonparallel lines.
 - **Infinitely many solutions.** In $\mathbb{R}^2$, the linear equations would be parallel, overlapping lines.

If a system of linear equations has a **solution set**, it is **consistent**. Otherwise, it is **inconsistent**.

Matrix Notation

We can display a system of linear equations with a matrix as follows.

$$\begin{array}{rcl} 5x_1 + x_2 - 9x_3 & = & 2 \\ x_1 & + & 4x_3 = 7 \\ & & 6x_3 = 1 \end{array} \qquad \left(\begin{array}{ccc|c} 5 & 1 & -9 & 2 \\ 1 & 0 & 4 & 7 \\ 0 & 0 & 6 & 1 \end{array} \right)$$

- The matrix above is a **coefficient matrix**.
- Each column corresponding to a different variable.
- An **augmented matrix** has an augmented column of the system's constants, usually separated by a line for clarity.

$$\left(\begin{array}{ccc|c} 5 & 1 & -9 & 2 \\ 1 & 0 & 4 & 7 \\ 0 & 0 & 6 & 1 \end{array} \right)$$

Solving a Linear System

To solve a linear system, we can replace it with an equivalent system. We can create an equivalent system by applying an **elementary row operation** to the rows (or in equation form, equations).

Definition 1.1.1 (Elementary Row Operations)

- 1) **Replacement:** Replace one row by the sum of itself and a scalar multiple of another row.
- 2) **Interchange:** Swap two rows.
- 3) **Scaling:** Multiplies all entries of a row by a nonzero constant.



- Thus, two matrices are **row equivalent** if there exists a sequence of elementary row operations that transforms one into the other.
- All elementary row operations are reversible.

1.2 Row Reduction and Echelon Forms

Definition 1.2.1 (Echelon Form)

A matrix is in **echelon form**, also known as **row echelon form**, if it has the following properties:

- 1) All nonzero rows are above any rows of all zeros.
- 2) Each leading (first) entry of a row is in a column to the right of the leading entry of the row above it.
- 3) All entries (in the same column) below a leading entry are zeros.



Definition 1.2.2 (Reduced Row Echelon Form)

Furthermore, a matrix is in **reduced row echelon form (rref)** if it satisfies the previous three properties and the following properties:

- 1) The leading entry in each row is 1.
- 2) Each leading 1 is the only nonzero entry in its column.



Theorem 1.2.3 (Uniqueness of the Reduced Echelon Form)

Each matrix is row equivalent to one reduced echelon form matrix.



- Because the leading values of an echelon form matrix can be anything, they are not unique.

Pivot Positions

- A **pivot position** in a matrix is the location that corresponds to the location of a leading 1 in the matrix's reduced row echelon form.
- A **pivot column** is any column containing a pivot position.

The Row Reduction Algorithm

Algorithm 1.2.4 (Row Reduction Algorithm)

- 1) Begin with the leftmost nonzero column. This will be a pivot column with a pivot position at the top.
- 2) Select a nonzero entry in the pivot column as the pivot. If needed, swap rows to move it into the pivot position.
- 3) Use elementary row operations to turn all entries below the pivot into zeros.
- 4) Repeat steps (1-3) with each consecutive nonzero column with pivot positions one row below the previous one.
- 5) Starting with the rightmost pivot, using scaling, if necessary, to convert each pivot to a 1.



- When performing an elementary row operation, we refer to each n^{th} row as R_n .

$$\begin{pmatrix} 0 & 8 \\ 3 & 5 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \end{matrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 3 & 5 \\ 0 & 8 \end{pmatrix} \xrightarrow{R_1 - \frac{5}{8}R_2 \rightarrow R_1} \begin{pmatrix} 3 & 0 \\ 0 & 8 \end{pmatrix} \xrightarrow{\frac{1}{3}R_1 \rightarrow R_1} \begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} \xrightarrow{\frac{1}{8}R_2 \rightarrow R_1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Solutions of Linear Systems

Applying the row reduction algorithm to an augmented matrix leads directly to a solution set for the associated linear system.

- **Basic variables** are variables whose coefficients are in a pivot column.
- **Free variables** are variables whose coefficients aren't in a pivot column.

If a system is consistent and has only one solution, then there will be no free variables.

If a system is consistent and has infinitely many solutions, then a solution set can be found by *finding the basic variables in terms of the free variables*.

If a system is inconsistent, then the solution set is empty.

Existence and Uniqueness

Theorem 1.2.5 (Existence and Uniqueness Theorem)

A linear system is consistent $\Leftrightarrow$ the corresponding augmented matrix's rightmost column is *not* a pivot column.

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.



1.3 Vector Equations

A matrix with one column is known as a **column vector**, or just a **vector**.

- Vectors are generally denoted using boldface lowercase letters:

$$\mathbf{u} = \begin{pmatrix} 5 \\ 2 \\ 6 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 19 \\ -5 \end{pmatrix} \quad \mathbf{w} = \begin{pmatrix} -2 \\ 0 \\ 23 \\ -9 \end{pmatrix}$$

- Sometimes, vectors are used to express columns of matrix.

In general, the set of all vectors across some n -dimensional coordinate space is $\mathbb{R}^n$. We will often work with vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ when geometric interpretations are needed, but going beyond the third dimension is not uncommon.

Geometric Vector Descriptions

For vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$, we can interpret vectors $\mathbf{u} = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ as directed line segments from the origin to the points (a, b) and (a, b, c) , respectively.

Vectors in $\mathbb{R}^n$

Property 1.3.1 (Algebraic Properties of $\mathbb{R}^n$)

$\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and all scalars c and d :

- i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
- iii) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
- iv) $\mathbf{u} + (-\mathbf{u}) = -(\mathbf{u}) + \mathbf{u} = \mathbf{0}$
- v) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
- vi) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
- vii) $c(d\mathbf{u}) = (cd)\mathbf{u}$
- viii) $1\mathbf{u} = \mathbf{u}$

Linear Combinations

A **linear combination** of $\mathbb{R}^n$ vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ is defined as follows:

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p$$

where scalars $c_1, c_2, \dots, c_p$ are the **weights**.

- We can express infinite sets of vectors as linear combinations of finite sets of vectors.
- Infinitely large solution sets are expressed using linear combinations.

Definition 1.3.2 (Span)

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ are vectors in $\mathbb{R}^n$, then the set of all linear combinations of those vectors is denoted by $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$** .

- The question of whether or not some vector $\mathbf{b}$ is in $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is equivalent to whether or not the following vector equation has a solution:

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{b}$$

- Relating this back to chapter 1.2, this is equivalent to this linear system:

$$(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_p)$$

Geometric Span Descriptions

Let $\mathbf{u}$ and $\mathbf{v}$ be some nonzero vectors in $\mathbb{R}^3$.

- $\text{Span}\{\mathbf{u}\}$ is a line straight through $\mathbf{0}$ and $\mathbf{u}$.
- $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is a plane that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{0}$ lie on.

These interpretations stem from the spans containing *all linear combinations of the vectors* in these subsets of $\mathbb{R}^3$.

1.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is a vector in $\mathbb{R}^n$, then the product $A\mathbf{x}$ is some vector $\mathbf{b} \in \mathbb{R}^m$ given by

$$A\mathbf{x} = (\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

- The form $A\mathbf{x} = \mathbf{v}$ is known as the **matrix equation**, as they can be analogous to linear systems.
- Matrices are often denoted by uppercase letters and are sometimes bolded.

Existence of Solutions

Theorem 1.4.1

Let A be an $m \times n$ matrix. Then, the following statements are equivalent.

- $\forall \mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
- Each $\mathbf{b} \in \mathbb{R}^m$ is a linear combination of the columns of A .
- The columns of A span $\mathbb{R}^m$.
- A has a pivot position in every row.

Caution

The matrix A of the matrix equation is a coefficient matrix, not an augmented matrix.

Properties of the Matrix-Vector Product

Theorem 1.4.2

If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then:

- $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$
- $A(c\mathbf{u}) = c(A\mathbf{u})$

1.5 Solution Sets of Linear Systems

Homogenous Linear Systems

A linear system is **homogenous** if it can be written in the form $A\mathbf{x} = \mathbf{0}$ given some $m \times n$ matrix A and the $\mathbb{R}^m$ zero vector $\mathbf{0}$.

- A homogenous system will always have at least one solution, specifically the **trivial solution**, $\mathbf{x} = \mathbf{0}$.
- Generally, we are interested in whether or not a homogenous system has a **nontrivial solution**.

Proposition 1.5.1

The homogenous equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\Leftrightarrow$ the equation has at least one free variable.

Example 1.5.2

Solve the homogenous linear system below:

$$8x_1 + 4x_2 + 5x_3 = 0$$

$$2x_1 + 4x_2 = 0$$

$$3x_1 + 2x_2 + x_3 = 0$$

$$\left(\begin{array}{ccc|c} 8 & 4 & 5 & 0 \\ 2 & 4 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array}\right) \xrightarrow{R_2 - \frac{1}{4}R_1 \rightarrow R_2} \left(\begin{array}{ccc|c} 8 & 4 & 5 & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 3 & 2 & 1 & 0 \end{array}\right) \xrightarrow{R_3 - \frac{3}{8}R_1 \rightarrow R_3} \left(\begin{array}{ccc|c} 8 & 4 & 5 & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & -\frac{1}{2} & -\frac{7}{8} & 0 \end{array}\right)$$

$$\xrightarrow{R_1 - \frac{4}{3}R_2 \rightarrow R_1} \left(\begin{array}{ccc|c} 8 & 0 & \frac{20}{3} & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & -\frac{1}{2} & -\frac{7}{8} & 0 \end{array}\right) \xrightarrow{R_3 + \frac{1}{6}R_2 \rightarrow R_3} \left(\begin{array}{ccc|c} 8 & 0 & \frac{20}{3} & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & 0 & -\frac{13}{12} & 0 \end{array}\right)$$

$$\xrightarrow{-\frac{12}{13}R_3 \rightarrow R_3} \left(\begin{array}{ccc|c} 8 & 0 & \frac{20}{3} & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{R_1 - \frac{20}{3}R_3 \rightarrow R_1} \left(\begin{array}{ccc|c} 8 & 0 & 0 & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow{R_2 + \frac{5}{4}R_3 \rightarrow R_2} \left(\begin{array}{ccc|c} 8 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{\frac{1}{8}R_1 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{\frac{1}{3}R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

Because there exists no free variables, the only solution is the trivial solution, $\mathbf{x} = \mathbf{0}$.

Parametric Vector Form

An *explicit description* of a solution set is often written in parametric vector form:

$$\mathbf{x} = s\mathbf{u} + t\mathbf{v}$$

where s and t are some real numbers.

Example 1.5.3

Describe the solutions of $A\mathbf{x} = \mathbf{0}$ in parametric vector form given $A = \begin{pmatrix} 1 & 2 & -2 & 5 \\ 0 & 1 & -5 & 3 \end{pmatrix}$.

$$\left(\begin{array}{cccc|c} 1 & 2 & -2 & 5 & 0 \\ 0 & 1 & -5 & 3 & 0 \end{array} \right) \xrightarrow{R_1 - 2R_2 \rightarrow R_1} \left(\begin{array}{cccc|c} 1 & 0 & 8 & -1 & 0 \\ 0 & 1 & -5 & 3 & 0 \end{array} \right)$$

$$\therefore x_1 + 8x_3 - x_4 = 0$$

$$x_2 - 5x_3 + 3x_4 = 0$$

$$\mathbf{x} = x_3 \begin{pmatrix} -8 \\ 5 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

Solutions of Nonhomogeneous Systems

When a nonhomogeneous linear system has infinitely many solutions, the general solution can be expressed in parametric vector form as *one solution to the system plus the general solution to the analogous homogeneous linear system*.

- Geometrically, solution sets between corresponding homogeneous and nonhomogeneous linear systems are *parallel*. Essentially, the solution set for any $A\mathbf{x} = \mathbf{b}$ is translated by $\mathbf{b}$ from the solution set for $A\mathbf{x} = \mathbf{0}$.

1.6 Applications of Linear Systems

1.7 Linear Independence

Definition 1.7.1

An indexed set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0}$$

only has the trivial solution. Otherwise, the set is **linearly dependent**.



Linear Independence of Matrix Columns

Notice how the definition for the linear dependence is analogous to the matrix equation $A\mathbf{x} = \mathbf{0}$. Recognizing each column of A as a vector, we can say that if $A\mathbf{x} = \mathbf{0}$ only has the trivial solution, *then its columns are linearly independent*.

- If we want to test the linear dependence of a set of vectors, we can let them be the columns of a matrix, then solve the corresponding homogeneous system.
- A set of one vector is linearly independent $\Leftrightarrow$ the vector is not the zero vector.

Sets of Two or More Vectors

Theorem 1.7.2 (Characterization of Linearly Dependent Sets)

An indexed set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ of at least two vectors is linearly dependent $\Leftrightarrow$ at least one of the vectors in S is a linear combination of the others.

If S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then there exists some $\mathbf{v}_j \forall j > 1$ such that $\mathbf{v}_j$ is a linear combination of the previous vectors.



1.8 Linear Transformations

A **transformation**, **function**, or **mapping** T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that maps each vector $\mathbf{x} \in \mathbb{R}^n$ to a vector $T(\mathbf{x}) \in \mathbb{R}^m$.

- This mapping is generally denoted $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$.
- $\mathbb{R}^n$ is the **domain**.
- $\mathbb{R}^m$ is the **codomain**.
- $T(\mathbf{x})$ is the **image** of $\mathbf{x}$ under T .
- The set of all images $T(\mathbf{x})$ is the **range**.

Matrix Transformations

Many transformations are associated with matrix multiplication. A **matrix transformation** is generally denoted $\mathbf{x} \mapsto A\mathbf{x}$ where A is the **standard matrix** of the transformation. If this matrix transformation is $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, then $T(\mathbf{x}) = A\mathbf{x}$.

- In this case, because the domain is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$, the standard matrix A must be $m \times n$.

Linear Transformations

Definition 1.8.1 (Linear Transformation)

A transformation T is **linear** if

- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \forall \mathbf{u}, \mathbf{v}$ in the domain of T .
- $T(c\mathbf{u}) = cT(\mathbf{u}) \forall$ scalars c and $\mathbf{u}$ in the domain of T .

- Essentially, a linear transformation requires a standard matrix A , as these properties stem from the properties of mapping $\mathbf{x} \mapsto A\mathbf{x}$:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$$

$$A(c\mathbf{u}) = cA\mathbf{u}$$

As a result, we can also ascertain two more properties of linear transformations.

- Given a linear transformation T , scalars c and d , and vectors $\mathbf{u}$ and $\mathbf{v}$ in the T 's domain:

$$T(\mathbf{0}) = \mathbf{0}$$

$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$

1.9 The Matrix of a Linear Transformation

Earlier, we asserted that every linear transformation requires some standard matrix A . We can prove this using identity matrix and **standard basis vectors** $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ (the columns of I , essentially).

Theorem 1.9.1

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then, there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

In fact, A is an $m \times n$ matrix whose j^{th} column is $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j^{th} column of I_n .

$$A = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix}$$

Proof.

$$T(\mathbf{x}) = T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n)$$

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n)$$

$$T(\mathbf{x}) = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

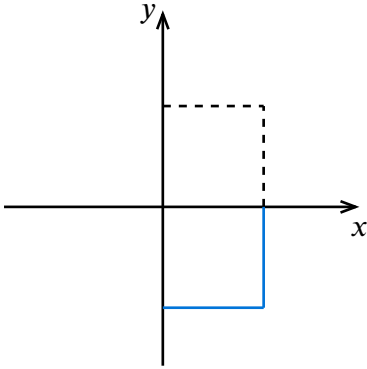
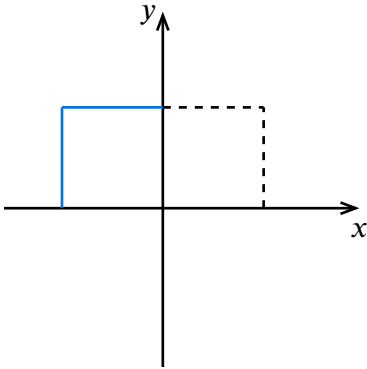
$$T(\mathbf{x}) = A\mathbf{x}$$

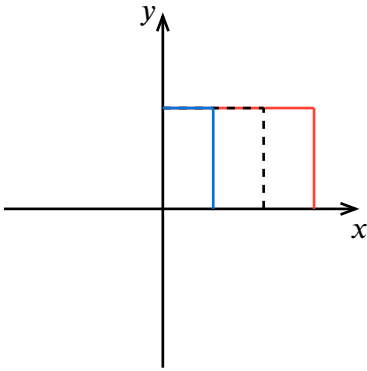
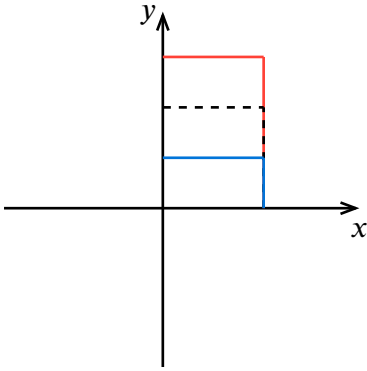
□

- Essentially, we can define any standard matrix for a linear transformation based on what they do to standard basis vectors.

Geometric Linear Transformations of $\mathbb{R}^2$

In $\mathbb{R}^2$, let's imagine $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ as the unit square. This will make it easier to interpret the following linear transformations geometrically.

Reflections		
Transformation	Unit Square Image	Standard Matrix
Reflection over the x -axis.		$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection over the y -axis.		$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Contractions and Expansions		
Transformation	Unit Square Image	Standard Matrix
Horizontal contraction and expansion.		$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$
Vertical contraction and expansion.		$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$

One-to-One and Onto

Definition 1.9.2

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** or **surjective** if each $\mathbf{b} \in \mathbb{R}^m$ is the image of *at least one* $\mathbf{x} \in \mathbb{R}^n$.



- Equivalently, T is onto if its codomain and range are equal.

Definition 1.9.3

A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** or **injective** if each $\mathbf{b} \in \mathbb{R}^m$ is the image of *at most one* $\mathbf{x} \in \mathbb{R}^n$.



- Equivalently T is one-to-one if $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or no solution.

When a mapping is both onto *and* one-to-one, it is **bijective**.

2 Matrix Algebra

2.1 Matrix Operations

Scalar multiples of matrices and matrix addition is done by component. Thus, matrices may only be added together if they have the same size.

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix}$$

$$A + B = \begin{pmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{pmatrix}$$

Theorem 2.1.1

Let A , B , and C be $m \times n$ matrices, and let r and s be scalars.

- a) $A + B = B + A$
- b) $(A + B) + C = A + (B + C)$
- c) $A + 0 = A$
- d) $r(A + B) = rA + rB$
- e) $(r + s)A = rA + sA$
- f) $r(sA) = (rs)A$



Matrix multiplication is defined similarly to matrix-vector multiplication. This hinges on the fact that $(AB)\mathbf{x} = A(B\mathbf{x})$ given that A is $m \times n$, B is $n \times p$, and $\mathbf{x} \in \mathbb{R}^p$.

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = (\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n) \quad B = \begin{pmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{np} \end{pmatrix} = (\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_p)$$

$$AB = (\mathbf{A}\mathbf{b}_1 \ \mathbf{A}\mathbf{b}_2 \ \cdots \ \mathbf{A}\mathbf{b}_p)$$

- In this case, AB is a valid product because the number of columns in A is equal to the number of rows in B . The resulting matrix is $m \times p$.

Caution

Matrix multiplication is not necessarily commutative.

Properties of Matrix Multiplication

Theorem 2.1.2

Let A be an $m \times n$ matrix, and let B and C have valid sizes for the indicated sums and products.

- a) $A(BC) = (AB)C$
- b) $A(B + C) = AB + AC$
- c) $(B + C)A = BA + CA$
- d) $r(AB) = (rA)B = A(rB)$ for any scalar r
- e) $I_m A = A I_n = A$



The Transpose of a Matrix

Given an $m \times n$ matrix A , the **transpose** of A , denoted A^T , is an $n \times m$ matrix whose columns are formed from the corresponding rows in A .

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \qquad A^T = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}$$

Theorem 2.1.3

Let A and B be matrices with valid sizes for the following sums and products.

- a) $(A^T)^T = A$
- b) $(A + B)^T = A^T + B^T$
- c) $(rA)^T = rA^T$ for any scalar r
- d) $(AB)^T = B^T A^T$



2.2 The Inverse of a Matrix

An $n \times n$ (**square**) matrix A is **invertible** or **nonsingular** if there exists an $n \times n$ matrix C such that

$$CA = AC = I$$

where C is the **inverse** of A . Thus, C may also be denoted as A^{-1} .

- Due to the nature of matrix multiplication, rectangular matrices can be **left invertible** or **right invertible**, but *not both*.

Theorem 2.2.1

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

- The quantity $ad - bc$ is the **determinant** of A , denoted $\det A$. Determinants of larger matrices will be covered in chapter 3.

Theorem 2.2.2

If A is an invertible $n \times n$ matrix, then $\forall \mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

Proof. Suppose A is any invertible $n \times n$ matrix and then $\mathbf{b}$ is some vector in $\mathbb{R}^n$.

$$A\mathbf{x} = \mathbf{b}$$

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$$

$$I\mathbf{x} = A^{-1}\mathbf{b}$$

$$\mathbf{x} = A^{-1}\mathbf{b}$$

□

Theorem 2.2.3

Let A and B be invertible matrices.

a) $(A^{-1})^{-1} = A$

b) $(AB)^{-1} = B^{-1}A^{-1}$

c) $(A^T)^{-1} = (A^{-1})^T$

Proof. Suppose that A and B are invertible matrices.

$$A^{-1}(A^{-1})^{-1} = I$$

$$(A^{-1})^{-1} = AI$$

$$(A^{-1})^{-1} = A$$

$$\begin{aligned}
AB(AB)^{-1} &= I \\
B(AB)^{-1} &= A^{-1}I \\
(AB)^{-1} &= B^{-1}A^{-1}I \\
(AB)^{-1} &= B^{-1}A^{-1}
\end{aligned}$$

$$\begin{aligned}
(A^{-1})^T A^T &= (AA^{-1})^T & A^T (A^{-1})^T &= (A^{-1}A)^T \\
(A^{-1})^T A^T &= I^T & A^T (A^{-1})^T &= I^T \\
(A^{-1})^T A^T &= I & A^T (A^{-1})^T &= I
\end{aligned}$$

□

Elementary Matrices

An **elementary matrix** is a matrix obtained from performing a single elementary row operation on an identity matrix, generally denoted E .

$$\begin{aligned}
E_1 &= \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & E_2 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} & E_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{pmatrix} \\
R_1 - 5R_3 &\rightarrow R_1 & R_2 \leftrightarrow R_3 & 9R_3 \rightarrow R_3
\end{aligned}$$

- They are particularly useful in proofs for representing elementary row operations, as left multiplying by an elementary matrix corresponds to one elementary row operation.
- Because elementary row operations are reversible, elementary matrices are invertible.

Theorem 2.2.4

An $n \times n$ matrix A is invertible $\Leftrightarrow A$ is row equivalent to I_n . In other words, there exists a sequence of elementary row operations that reduces A to I_n . Additionally, that sequence also reduces I_n to A^{-1} .

♡

Proof. Suppose A is an invertible $n \times n$ matrix. Then, $\forall \mathbf{b} \in \mathbb{R}^n$, there exists a unique solution to $\mathbf{x} = A^{-1}\mathbf{b}$ to $A\mathbf{x} = \mathbf{b}$. Because A is square, the main diagonal represents its pivot positions, implying that A can be reduced to I_n .

Now, suppose that A is row equivalent to I_n . Thus, there exists a sequence of elementary row operations represented by elementary matrices $E_1, E_2, \dots, E_p$ that reduce A to I_n . Because elementary matrices are invertible, $E_p \cdots E_2 E_1$ is invertible. By definition, $E_p \cdots E_2 E_1$ is an inverse for A . Thus, A is invertible. Additionally:

$$\begin{aligned}
E_p \cdots E_2 E_1 A &= I_n \\
E_p \cdots E_2 E_1 A A^{-1} &= I_n A^{-1} \\
E_p \cdots E_2 E_1 I_n &= A^{-1}
\end{aligned}$$

□

An Algorithm for A^{-1}

We can augment any $n \times n$ matrix A with I , then reduce A , if possible, to I . By the previous theorem, it follows that the augmented columns will form A^{-1} once A is reduced to I .

$$(A \ I) \sim (I \ A)$$

Example 2.2.5

Find the inverse of $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}$, if it exists.

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3 - 4R_1 \rightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array} \right) \\ & \xrightarrow{R_3 + 3R_2 \rightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array} \right) \xrightarrow{\frac{1}{2}R_3 \rightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right) \\ & \xrightarrow{R_1 - 3R_3 \rightarrow R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right) \xrightarrow{R_2 - 2R_3 \rightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & -4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right) \\ & A^{-1} = \begin{pmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & -4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} \end{aligned}$$

◇

2.3 Characteristics of Invertible Matrices

Theorem 2.3.1 (Invertible Matrix Theorem)

Let A be an $n \times n$ matrix. The following statements are equivalent:

- A is invertible.
- A is row equivalent to I_n .
- A has n pivot positions.
- The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
 - The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
 - There exists some $n \times n$ matrix C such that $CA = I$.
 - There exists some $n \times n$ matrix D such that $AD = I$.
 - A^T is invertible.

Note

In later chapters, we will make multiple amendments to this theorem.

Invertible Linear Transformations

Say the mapping T is defined by $\mathbf{x} \mapsto A\mathbf{x}$. If A is invertible, then we could define an **inverse function** $S = T^{-1}$ as $\mathbf{x} \mapsto A^{-1}\mathbf{x}$. Below, we can show that these mappings are indeed inverses.

$$S(T(\mathbf{x})) = A^{-1}(A\mathbf{x}) = (A^{-1}A)\mathbf{x} = I\mathbf{x} = \mathbf{x}$$

$$T(S(\mathbf{x})) = A(A^{-1}\mathbf{x}) = (AA^{-1})\mathbf{x} = I\mathbf{x} = \mathbf{x}$$

Theorem 2.3.2

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then, T is invertible $\Leftrightarrow A$ is invertible. In that case, the linear transformation S defined by $S(\mathbf{x}) = A^{-1}\mathbf{x}$ is a unique mapping that satisfies the following equations:

$$S(T(\mathbf{x})) = \mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

$$T(S(\mathbf{x})) = \mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

2.4 Matrix Factorizations

A factorization of a matrix expresses the matrix as a product of two or more matrices. Often, specific factorizations can make specific matrix computations far less intensive.

LU Factorization

- A **lower triangular matrix** is a square matrix with solely zero entries above the main diagonal.
 - A **unit lower triangular matrix** has only 1s along the main diagonal.

Proposition 2.4.1

Products and inverses of unit lower triangular matrices are also unit lower triangular matrices.

- An **upper triangular matrix** is a square matrix with solely zero entries below the main diagonal. The echelon form of an $n \times n$ matrix is an example of this.

If A can be reduced to an echelon form U by a sequence of row replacements corresponding to unit lower triangular matrices $E_1, E_2, \dots, E_p$, then $A = (E_p \cdots E_2 E_1)U$. Thus, we can factorize A as LU if $L = (E_p \cdots E_2 E_1)^{-1}$.

Using this factorization to solve $A\mathbf{x} = \mathbf{b}$, we get $(LU)\mathbf{x} = L(U\mathbf{x}) = \mathbf{b}$.

- If we let $U\mathbf{x} = \mathbf{y}$, then we can solve for $\mathbf{x}$ by first solving $L\mathbf{y} = \mathbf{b}$, then substituting into that equation.
- For larger matrices, constructing L and U take far less time than reducing the original matrix to row reduced echelon form.

Algorithm 2.4.2 (LU Factorization)

- 1) Reduce A to an echelon form U by a sequence of elementary row operations, if possible.
- 2) Place entries in L such that the *same sequence* of row operations reduces L to I .

For each pivot column in U , let the pivot and the entries below the pivot be a column in L , then apply row replacements to U such that the entries below the pivot become zeros.

- Now, by proposition (2.4.1), L is a unit lower triangular matrix.
- Thus, when copying U 's pivot columns onto L , let all entries in each column above the pivot be 0.
- Additionally, divide each column by its pivot.

Note

While L will always be a square matrix, A (and by extension, U) do not have to be square matrices.

Example 2.4.3

Solve $A\mathbf{x} = \mathbf{b}$ using an LU factorization given $A = \begin{pmatrix} 3 & -5 & 3 \\ -9 & 12 & -4 \\ 9 & -12 & 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 15 \\ -15 \end{pmatrix}$.

$$\begin{pmatrix} 3 & -5 & 3 \\ -9 & 12 & -4 \\ 9 & -12 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} 3 & -5 & 3 \\ 0 & -3 & 5 \\ 0 & 3 & -4 \end{pmatrix} \rightarrow \begin{pmatrix} 3 & -5 & 3 \\ 0 & -3 & 5 \\ 0 & 0 & 1 \end{pmatrix} = U$$

$$\begin{pmatrix} 3 & 0 & 0 \\ -9 & -3 & 0 \\ 9 & 3 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 3 & -1 & 1 \end{pmatrix} = L$$

$$L\mathbf{y} = \mathbf{b}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ -3 & 1 & 0 & 15 \\ 3 & -1 & 1 & -15 \end{array} \right) \xrightarrow{R_2+3R_1 \rightarrow R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 12 \\ 3 & -1 & 1 & -15 \end{array} \right) \xrightarrow{R_3-3R_1 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 12 \\ 0 & -1 & 1 & -12 \end{array} \right)$$

$$\xrightarrow{R_3+R_2 \rightarrow R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 12 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$\therefore \mathbf{y} = \begin{pmatrix} -1 \\ 12 \\ 0 \end{pmatrix}$$

$$U\mathbf{x} = \mathbf{y}$$

$$\left(\begin{array}{ccc|c} 3 & -5 & 3 & -1 \\ 0 & -3 & 5 & 12 \\ 0 & 0 & 1 & 0 \end{array} \right) \xrightarrow{R_1-3R_3 \rightarrow R_1} \left(\begin{array}{ccc|c} 3 & -5 & 0 & -1 \\ 0 & -3 & 5 & 12 \\ 0 & 0 & 1 & 0 \end{array} \right) \xrightarrow{R_2-5R_3 \rightarrow R_2} \left(\begin{array}{ccc|c} 3 & -5 & 0 & -1 \\ 0 & -3 & 0 & 12 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$\xrightarrow{-\frac{1}{3}R_2 \rightarrow R_2} \left(\begin{array}{ccc|c} 3 & -5 & 0 & -1 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array} \right) \xrightarrow{R_1+5R_2 \rightarrow R_1} \left(\begin{array}{ccc|c} 3 & 0 & 0 & -21 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$\xrightarrow{R_1+5R_2 \rightarrow R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$\therefore \mathbf{x} = \begin{pmatrix} -7 \\ -4 \\ 0 \end{pmatrix}$$

◇

2.5 Homogeneous Coordinates

The **homogeneous coordinates** of a point $(x, y) \in \mathbb{R}^2$ is $(x, y, 1) \in \mathbb{R}^3$.

- Recall how linear transformations must map $\mathbf{0}$ to $\mathbf{0}$, meaning that we were unable to express translations.
- Using homogeneous coordinates, we can translate vectors by $\begin{pmatrix} h \\ k \end{pmatrix}$ on the plane $z = 1$ using the following standard matrix:

$$A = \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

- The other geometric $\mathbb{R}^2$ linear transformations we covered in (1.9) are expressed with standard matrices similar to their previous ones. For example:

Rotation

$$A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Reflection over vertical axis

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Homogeneous 3D Coordinates

In general, the homogeneous coordinates of a point $(x, y, z) \in \mathbb{R}^3$ is $\left(\frac{x}{H}, \frac{y}{H}, \frac{z}{H}, H\right) \in \mathbb{R}^4$ where $h \neq 0$ and $x = \frac{X}{H}$, $y = \frac{Y}{H}$, and $z = \frac{Z}{H}$.

- Thus, nonzero scalar multiples of $(x, y, z, 1)$ would also be homogeneous coordinates for (x, y, z) .

Perspective Projections

Imagine the xy -plane as a viewing frame for a scene, and imagine that a viewer observes from some point $(0, 0, d)$ on the positive z -axis. A **perspective projection** maps each point (x, y, z) onto an *image point* $(x^*, y^*, 0)$ such that those points and the viewing position lie on a single line.

- Drawing out these lines, we find the image's coordinates in terms of the original position and the viewing position.

$$\frac{x^*}{d} = \frac{x}{d-z} \Rightarrow x^* = \frac{dx}{d-z} = \frac{x}{1-z/d}$$

$$\frac{y^*}{d} = \frac{y}{d-z} \Rightarrow y^* = \frac{dy}{d-z} = \frac{y}{1-z/d}$$

- Applying homogeneous coordinates, we are mapping $(x, y, z, 1)$ to $\left(\frac{x}{1-z/d}, \frac{y}{1-z/d}, 0, 1\right)$, or equivalently, $\left(x, y, 0, 1 - \frac{z}{d}\right)$.

Now, we can create a general 4×4 matrix P for all $\mathbb{R}^3$ perspective projections.

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{d} & 1 \end{pmatrix}$$

$$P \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{d} & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \\ 1 - \frac{z}{d} \end{pmatrix} \equiv \begin{pmatrix} \frac{dx}{d-z} \\ \frac{dy}{d-z} \\ 0 \\ 1 \end{pmatrix}$$

3 Determinants

In chapter (2.2), we defined determinants for 2×2 matrices. In this chapter, we will cover the general definition for $n \times n$ matrices.

3.1 Introduction to Determinants

Definition 3.1.1

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$.

$$\begin{aligned} \det A &= a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n} \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j} \end{aligned}$$

A_{ij} denotes the **(i, j) -minor** of A . That is, the matrix obtained from removing the i^{th} row and j^{th} column from A .

- The determinant of some $n \times n$ matrix $A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$ can be expressed using vertical lines in lieu of the parentheses (or brackets).

$$\det A = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}$$

The **(i, j) -cofactor** of some matrix A , denoted C_{ij} , is denoted as follows:

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

A **cofactor expansion** is a method of calculating a determinant using any row or column. For instance, our definition for determinant uses a *cofactor expansion across the first row*.

Theorem 3.1.2

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column.

The cofactor expansion across the i^{th} row of A is

$$\det A = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}$$

The cofactor expansion down the j^{th} column of A is

$$\det A = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj}$$

- Because we can choose any row or column to do a cofactor expansion along, choosing rows or columns with more zeros is preferable.

 Note

The sign of $(-1)^{i+j}$ in every cofactor follows the same checkerboard pattern for every matrix.

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Theorem 3.1.3

If A is triangular or diagonal matrix, then $\det A$ is a product of the entries along the main diagonal of A .

$$\det A = \prod_{i=1}^n a_{ii}$$



Proof. Suppose U is an upper triangular matrix. If U is a 2×2 matrix, then

$$\det U = u_{11}u_{22} - u_{12} \cdot 0 = u_{11}u_{22}$$

Now, suppose that if U is $n \times n$, $\det U = u_{11}u_{22} \cdots u_{nn}$. Let U' be an $(n+1) \times (n+1)$ upper triangular matrix such that $U'_{11} = U$. If we do a cofactor expansion down the first column of U' , we get:

$$\det U' = u'_{11} \det U'_{11} - 0 \cdot \det U'_{21} + \cdots + (-1)^{n+1} \det U'_{n1}$$

$$\det U' = u'_{11} \det U$$

$$\det U' = u'_{11} (u_{11}u_{22} \cdots u_{nn})$$

$$\det U' = u'_{11} u'_{22} u'_{33} \cdots u'_{(n+1),(n+1)}$$

It follows that we can apply the same process to a diagonal matrix D , and apply a similar process to a lower triangular matrix L , albeit using the first column of each consecutive minor rather than the first row. □

 Important

Because the identity matrix is a diagonal matrix consisting solely of 1s, for any $n \times n$ identity matrix, $\det I_n = 1$.

3.2 Properties of Determinants

Row Operations

Theorem 3.2.1 (Row Operations)

Let A be an $n \times n$ matrix.

- If a row replacement is applied to A to produce B , then $\det B = \det A$.
- If two rows of A are swapped to produce B , then $\det B = -\det A$.
- If one row of A is scaled by some nonzero constant k , then $\det B = k \det A$.

Note

If an $n \times n$ matrix A is multiplied by some scalar k to produce B , then $\det B = k^n \det A$.

Theorem 3.2.2 (Invertible Matrix Theorem Amendment)

Let A be an $n \times n$ matrix. Then, the following statements are equivalent:

- A is invertible.
- ...
- $\det A \neq 0$

Column Operations

Theorem 3.2.3

If A is an $n \times n$ matrix, then $\det A^T = \det A$.

- Thus, column operations have equivalent effects on determinants as corresponding row operations.

Determinants and Matrix Product

Theorem 3.2.4 (Multiplicative Property)

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

- Thinking back to LU factorizations, we can also say that $\det A = \det L \det U$.

Linearity Property of the Determinant Function

For any $n \times n$ matrix A , $\det A$ is a function of the n column vectors of A . Now, suppose that all columns of A aside from the j^{th} column of A are held fixed such that we can define $\det A$ with transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}$ as follows:

$$T(\mathbf{x}) = \det(\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_{j-1} \ \mathbf{x} \ \mathbf{a}_{j+1} \ \cdots \ \mathbf{a}_{n-1} \ \mathbf{a}_n)$$

This is a linear transformation.

- Because all columns (aside from the j^{th} column) are held fixed, doing cofactor expansions along the j^{th} row shows the linearity properties:

$$\forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n \text{ and } \forall k \in \mathbb{R}$$

$$T(\mathbf{u} + \mathbf{v}) = (u_1 + v_1)C_{1j} + \cdots + (u_n + v_n)C_{nj}$$

$$T(\mathbf{u} + \mathbf{v}) = (u_1C_{1j} + \cdots + u_nC_{nj}) + (v_1C_{1j} + \cdots + v_nC_{nj})$$

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$$

$$T(k\mathbf{u}) = ku_1C_{1j} + \cdots + ku_nC_{nj}$$

$$T(k\mathbf{u}) = k(u_1C_{1j} + \cdots + u_nC_{nj})$$

$$T(k\mathbf{u}) = kT(\mathbf{u})$$

3.3 Cramer's Rule

Cramer's rule is used to solve for singular entries of a solution to $A\mathbf{x} = \mathbf{b}$. Because it involves determinants, it's very impractical for hand calculations. Rather, it is very useful for theoretical calculations or small matrices.

Theorem 3.3.1 (Cramer's Rule)

Let A be an $n \times n$ invertible matrix. For any $\mathbf{b} \in \mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}$$

where $1 \leq i \leq n$ and $A_i(\mathbf{b})$ denotes the matrix obtained from replacing the i^{th} column of A with $\mathbf{b}$.

Proof. Let A be an invertible $n \times n$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$.

$$A(I_i(\mathbf{x})) = A(\mathbf{e}_1 \cdots \mathbf{e}_{i-1} \ \mathbf{x} \ \mathbf{e}_{i+1} \cdots \mathbf{e}_n)$$

$$A(I_i(\mathbf{x})) = (A\mathbf{e}_1 \cdots A\mathbf{e}_{i-1} \ A\mathbf{x} \ A\mathbf{e}_{i+1} \cdots A\mathbf{e}_n)$$

$$A(I_i(\mathbf{x})) = (\mathbf{a}_1 \cdots \mathbf{a}_{i-1} \ \mathbf{b} \ \mathbf{a}_{i+1} \cdots \mathbf{a}_n)$$

$$A(I_i(\mathbf{x})) = A_i(\mathbf{b})$$

$$\therefore \det A(I_i(\mathbf{x})) = \det A_i(\mathbf{b})$$

$$\det A \det I_i(\mathbf{x}) = \det A_i(\mathbf{b})$$

$$\det I_i(\mathbf{x}) = \frac{\det A_i(\mathbf{b})}{\det A}$$

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}$$

□

A Formula for A^{-1}

Because the product of a matrix and its inverse equals I , the product of a matrix and the j^{th} column vector of its inverse equals the j^{th} unit basis vector. By Cramer's rule,

$$a_{ij}^{-1} = \frac{\det A_i(\mathbf{e}_j)}{\det A}$$

Now, we can simplify $\det A_i(\mathbf{e}_j)$ by cofactor expansion down the i^{th} column, because all entries in the column will be 0 aside from the j^{th} entry.

$$a_{ij}^{-1} = \frac{C_{ji}}{\det A}$$

Thus, if we do this for every entry of A^{-1} , we can create A^{-1} . Factoring out $\frac{1}{\det A}$ from each entry, each (i, j) entry corresponds to cofactor C_{ji} of A . This matrix is known as the **adjugate** of A .

$$\text{adj } A = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}$$

Theorem 3.3.2 (An Inverse Formula)

Let A be an invertible $n \times n$ matrix.

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

Determinants as Area or Volume

Theorem 3.3.3

If A is a 2×2 matrix, then the area of the parallelogram defined by the columns of A equals $|\det A|$.

If A is a 3×3 matrix, then the volume of the parallelepiped defined by the columns of A equals $|\det A|$.

Note

When using determinants to find area or volume, recall that column vectors should be from the origin. For example, a side between points $(1, 1)$ and $(5, 1)$ would correspond to the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$.

- We can interpret the determinant of a matrix as how much vectors are scaled under a corresponding transformation.

4 Vector Spaces

5 Eigen-everything

5.1 Eigenvectors and Eigenvalues

Definition 5.1.1 (Eigenvector and Eigenvalue)

An **eigenvector** of an $n \times n$ matrix A is any nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . A scalar λ is an **eigenvalue** of A if there exists a nontrivial solution $\mathbf{x}$ to $A\mathbf{x} = \lambda\mathbf{x}$ such that $\mathbf{x}$ is the eigenvector corresponding to λ .

- An **eigenspace** consists of all eigenvectors corresponding to some eigenvalue λ , and forms a subspace of $\mathbb{R}^n$.
- Geometrically, an eigenvalue is a scaling factor in a specific direction during a linear transformation.

Note

While eigenvectors cannot be zero, *eigenvalues can be zero*.

5.2 The Characteristic Equation

To solve for the eigenvectors, and by extension, the eigenvalues of a matrix, we can take advantage of the general matrix equation for eigenvectors.

$$\begin{aligned} A\mathbf{x} &= \lambda\mathbf{x} \\ A\mathbf{x} - \lambda\mathbf{x} &= 0 \\ (A - \lambda I_n)\mathbf{x} &= 0 \end{aligned}$$

- Now, by definition, the homogeneous equation above must have a nontrivial solution.
- Then, by the Invertible Matrix Theorem, $(A - \lambda I_n)$ is not invertible. Equivalently, its determinant is zero. This gives us the **characteristic equation**, a n^{th} degree polynomial whose roots are A 's eigenvalues.

Definition 5.2.1 (Characteristic Equation)

Let A be some $n \times n$ matrix. The characteristic equation is defined as follows:

$$\det(A - \lambda I_n) = 0$$

A scalar λ may only be an eigenvalue for A if it satisfies the equation above.

- We can use this to prove that for any $n \times n$ matrix A , A^T has the same eigenvalues.

Proof. Suppose A and A^T are some $n \times n$ matrices whose eigenvalues are given by $\det(A - \lambda I)$ and $\det(A^T - \lambda I)$, respectively.

$$\begin{aligned} \det(A^T - \lambda I) &= \det(A^T - \lambda I^T) \\ \det(A^T - \lambda I) &= \det(A - \lambda I) \end{aligned}$$

□

Similarity

If A and B are $n \times n$ matrices, then A is **similar to** B if there exists an invertible matrix P such that $A = PBP^{-1}$, or equivalently, $B = P^{-1}AP$.

- This relationship is *symmetric*.
- The mapping $A \mapsto P^{-1}AP$ is known as a **similarity transformation**.

Theorem 5.2.2

If two $n \times n$ matrices A and B are similar, then their characteristic polynomials, and by extension, their eigenvalues, are the same.

Proof. Let P be some $n \times n$ invertible matrix.

$$A - \lambda I = PBP^{-1} - \lambda PP^{-1}$$

$$A - \lambda I = P(B - \lambda)P^{-1}$$

$$\det(A - \lambda I) = \det(P(B - \lambda)P^{-1})$$

$$\det(A - \lambda I) = \det(P(B - \lambda)P^{-1})$$

$$\det(A - \lambda I) = \det(P) \det(B - \lambda) \det(P^{-1})$$

$$\det(A - \lambda I) = \det(P) \det(B - \lambda) \left(\frac{1}{\det(P)} \right)$$

$$\det(A - \lambda I) = \det(B - \lambda)$$

□

5.3 Diagonalization

A matrix is **diagonalizable** if it is similar to a diagonal matrix.

- This factorization is particularly helpful for calculating large k^{th} powers of matrices.

Theorem 5.3.1 (The Diagonalization Theorem)

An $n \times n$ matrix A is diagonalizable $\Leftrightarrow A$ has n linearly independent eigenvectors.

Specifically, $A = PDP^{-1}$ given diagonal matrix $D \Leftrightarrow$ the columns of P are n linearly independent eigenvectors of A and the diagonal entries of D are the eigenvalues corresponding to each i^{th} column in P .

- In other words, A is diagonalizable $\Leftrightarrow$ there exists enough eigenvectors to form an **eigenvector basis** of $\mathbb{R}^n$.

Proof. Let P be the a matrix whose columns are A 's eigenvectors, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and let D be a diagonal matrix whose diagonal entries are A 's eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Since the eigenvectors of A are linearly independent, the columns of P are linearly independent. By the Invertible Matrix Theorem, P is invertible. Since P , D , and P^{-1} are all well-defined, the diagonalization PDP^{-1} exists as follows:

$$\begin{aligned} AP &= A(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n) \\ AP &= (A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n) \\ AP &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n) \\ APP^{-1} &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n)P^{-1} \\ A &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n)P^{-1} \\ A &= PDP^{-1} \end{aligned}$$

Now, suppose A is diagonalizable such that $A = PDP^{-1}$. This is equivalent to $AP = PD$. Now, let P 's columns be vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and let D 's diagonal entries be $\lambda_1, \lambda_2, \dots, \lambda_n$.

$$AP = PD$$

$$\begin{aligned} A(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n) &= (\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n) \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \\ (A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n) &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n) \end{aligned}$$

□

5.4 Eigenvectors and Linear Transformations

5.5 Complex Eigenvalues

6 Orthogonality and Least Squares

Least

7 Symmetric Matrices and Quadratic Forms